

Slide 1 - AWS Cloud Foundations for Machine Learning

Hello everyone, and welcome to the first presentation in my AWS Certified Machine Learning Engineer - Associate learning series. This is BLA 1, Part 1, and the focus is AWS cloud foundations for machine learning. Before discussing models, algorithms, or generative AI, it is important to understand the environment in which AWS machine learning solutions operate. Machine learning on AWS is not limited to one service. It depends on cloud infrastructure, storage, compute, networking, security, monitoring, and managed AI services working together.

In this presentation, I will explain cloud computing in simple terms, introduce the AWS global infrastructure, organize important AWS services into functional layers, discuss the shared responsibility model, and connect these ideas to the AWS Well-Architected Framework. My goal is to create a foundation that will support every later presentation in this series. I am not demonstrating a lab or writing code. Instead, I am focusing on the reasoning behind AWS architecture and on the decisions a machine learning engineer must understand.

The key idea is that AWS gives organizations flexible access to technology resources, but flexibility does not remove responsibility. Engineers still have to choose the correct Region, services, security controls, capacity, and cost model. These choices affect whether a machine learning workload is reliable, secure, scalable, and affordable. By the end of this presentation, we should understand how the cloud changes the way machine learning systems are designed and why cloud knowledge is essential for the MLA-C02 certification.

Slide 2 - Cloud computing changes how ML resources are obtained

Cloud computing changes the way organizations obtain and manage technology resources. In a traditional environment, a company normally purchases physical servers, storage devices, networking equipment, and software licenses. It has to estimate future demand before that demand actually exists. If the company buys too little capacity, applications may become slow or unavailable. If it buys too much, expensive equipment may remain unused. The organization also needs space, power, cooling, hardware maintenance, and employees who can manage that infrastructure.

AWS uses a different operating model. Instead of purchasing all infrastructure in advance, teams can request computing resources when they need them. They can increase capacity when demand rises and reduce it when demand falls. AWS provides services through APIs, management consoles, command-line tools, and automation systems. Many services are managed, which means AWS takes responsibility for more of the underlying infrastructure and routine operations. Customers can therefore concentrate more attention on their data, applications, and business requirements.

This flexibility is especially valuable for machine learning. Training a model may require powerful compute resources for a limited period, while inference may need smaller resources that remain available continuously. A generative AI application may experience unpredictable traffic. A data-preparation job may process a large dataset once and then stop. AWS allows each workload to use a capacity model that fits its behavior.

However, cloud computing is not automatically cheaper or simpler. Poorly selected resources can create unnecessary cost, and incorrect configurations can create security or reliability problems. The important shift is that infrastructure becomes a set of services and decisions. A machine learning engineer must understand the workload, select appropriate services, monitor usage, and continuously balance performance with cost.

Slide 3 - AWS infrastructure separates location, scale, and resilience

AWS organizes its infrastructure into geographic Regions, Availability Zones, and edge locations. A Region is a separate geographic area where AWS operates infrastructure. Most AWS resources are created in a specific Region. When selecting a Region, an organization may consider where its users are located, whether a required service is available, how much that service costs, and whether laws or policies require data to remain in a certain location.

Each Region contains multiple Availability Zones. An Availability Zone consists of one or more data centers with independent power, networking, and connectivity. Availability Zones within a Region are connected, but they are designed so that a failure in one location is less likely to affect another. A workload that uses more than one Availability Zone can therefore be more resilient than a workload that depends on a single location. This matters for production machine learning endpoints because users may depend on predictions being available continuously.

Edge locations bring selected AWS capabilities closer to users. They are commonly associated with content delivery and low-latency access. They do not replace Regions, but they can improve how users interact with globally distributed applications. AWS also has services with a global scope. Identity and Access Management, for example, controls access across an AWS account even though many of the resources being accessed are regional.

For a machine learning engineer, infrastructure geography affects more than availability. It influences network latency, data-transfer cost, regulatory compliance, disaster recovery, and service selection. A dataset stored in one Region and a training job created in another may create additional complexity and expense. The certification therefore expects candidates to read a set of requirements and recognize which location and resilience strategy is appropriate rather than simply choosing a Region at random.

Slide 4 - An AWS ML solution combines four service layers

An AWS machine learning solution can be understood as four connected service layers: data, model development, deployment, and operations. This structure is useful because it organizes a large number of AWS services according to the role they perform. It also prevents us from thinking that a machine learning solution begins and ends with model training.

The first layer is data. Machine learning depends on information that may come from files, databases, applications, sensors, logs, or streaming platforms. Amazon S3 is widely used for scalable object storage. Services such as Amazon RDS, DynamoDB, Redshift, Kinesis, Glue, and EMR support different storage, ingestion, transformation, and analytics requirements. The correct service depends on data structure, volume, speed, access pattern, governance, and cost.

The second layer is model development. Amazon SageMaker AI supports traditional machine learning activities such as preparing data, training models, tuning hyperparameters, evaluating performance, and managing model versions. Amazon Bedrock provides managed access to foundation models and capabilities for generative AI, including knowledge bases, guardrails, evaluations, and agents. These services can support different approaches, so the engineer must select the one that fits the problem.

The third layer is deployment. A trained or customized model must deliver predictions or generated responses. AWS supports real-time endpoints, asynchronous inference, serverless inference, and batch processing. The fourth layer is operations. CloudWatch helps observe metrics and logs. IAM controls permissions. AWS KMS supports encryption key management, and CloudTrail records API activity. Cost-management and governance services help organizations control usage. Together, these layers form an end-to-end system in which every component affects the others.

Slide 5 - AWS secures the cloud; customers secure their workloads

The AWS shared responsibility model explains how security duties are divided between AWS and the customer. AWS describes its responsibility as security of the cloud. This includes protecting the physical data centers, hardware, networking, and foundational infrastructure that runs AWS services. Customers do not need to enter an AWS data center or repair a physical server because AWS manages those responsibilities.

The customer is responsible for security in the cloud. Customer responsibilities include protecting data, managing identities, assigning permissions, choosing encryption settings, configuring networks, securing applications, and reviewing activity. The exact boundary depends on the type of service. With a basic compute service, the customer manages more of the operating environment. With a highly managed service, AWS manages more of the underlying platform. Even then, the customer still decides which data is used, who can access it, and how the service should behave.

Machine learning introduces valuable and sometimes sensitive assets. Training data may contain personal information. Model artifacts may represent intellectual property. Endpoints can expose predictions to applications. Generative AI systems may process private prompts or return unsafe output if protections are missing. Logs can also contain sensitive details. Customers must use least-privilege permissions, encryption, network controls, monitoring, and appropriate data-governance practices.

The shared responsibility model is important on the exam because an answer that says AWS handles all security is incorrect. AWS provides secure infrastructure and security services, but customers have to configure and use those services correctly. A managed service reduces operational burden; it does not eliminate accountability. Understanding the division of responsibility helps an engineer identify which control belongs to AWS and which action must be performed by the organization using AWS.

Slide 6 - Good cloud decisions balance architectural qualities

A strong AWS design balances several architectural qualities rather than optimizing only one. The AWS Well-Architected Framework describes six pillars: operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability. These pillars provide a structured way to evaluate a workload and identify risks or tradeoffs.

Operational excellence focuses on running and improving systems effectively. For machine learning, this includes repeatable processes, useful monitoring, documented changes, and the ability to respond to incidents. Security focuses on protecting identities, data, models, endpoints, and infrastructure. Reliability means that a workload can perform its intended function and recover from failure. A reliable inference system should continue serving users or recover predictably when a component becomes unavailable.

Performance efficiency means selecting resources and architectures that meet requirements without wasting capacity. A real-time application may prioritize low latency, while a nightly prediction job may prioritize throughput and cost. Cost optimization requires understanding what resources are being consumed and selecting suitable pricing and scaling approaches. Sustainability considers how efficiently resources are used over time.

These qualities frequently create tradeoffs. A larger instance may improve performance but increase cost. Replicating resources across multiple Availability Zones may improve resilience but use more capacity. A managed service may reduce operational effort while providing a different level of control from a self-managed option. The correct architecture depends on the requirements stated in the scenario. MLA-C02 questions often test this decision process. Candidates must identify the most important requirement, eliminate options that violate it, and then select the design that provides the best overall balance.

Slide 7 - The cloud foundation gives us a decision framework

To conclude Part 1, the cloud foundation gives us a practical decision framework even though this presentation does not include a hands-on lab. The first question is where the workload should run. Region selection affects latency, compliance, service availability, resilience, and cost. The second question is which services provide the required capability. AWS offers many services, but the correct choice depends on the workload rather than on which service is most popular.

The third question is who manages each security responsibility. AWS protects the underlying cloud infrastructure, while customers remain responsible for their data, access decisions, configurations, and application behavior. The fourth question is how the design balances architectural qualities. Security, reliability, performance, operations, cost, and sustainability must be considered together.

The main takeaway is that AWS machine learning is an architectural system. Data services provide information. SageMaker AI and Bedrock support model development. Deployment services make model outputs available. Monitoring, identity, encryption, logging, and cost tools keep the solution controlled. The cloud makes these capabilities available on demand, but engineers still need to make informed decisions.

This foundation prepares us for Part 2, which focuses on machine learning concepts. We will examine classification, regression, clustering, and generative AI. We will compare traditional machine learning with foundation models, separate training from inference, discuss evaluation metrics, and explain underfitting, overfitting, bias, and drift. Those concepts will allow us to understand not only what AWS services do, but why an engineer would choose one approach over another. Thank you for watching Part 1 of BLA 1.